Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

ASSIGNMENT - 2

FACULTY : SWETA BHATTACHARYA

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REG NO : 23MIS0379

SUBJECT : AGILE DEVELOPMENT PROCESS AND DEVOPS

COURSE CODE: ISWE406P

1. Learning Goal

Understand what Jenkins is and run a job successfully.

Stepwise Tasks

1. What is CI/CD? (Real-world example)

CI (Continuous Integration): Developers frequently push code to a shared repository.

Jenkins automatically builds and tests the code.

CD (Continuous Delivery/Deployment): After successful testing, the application is

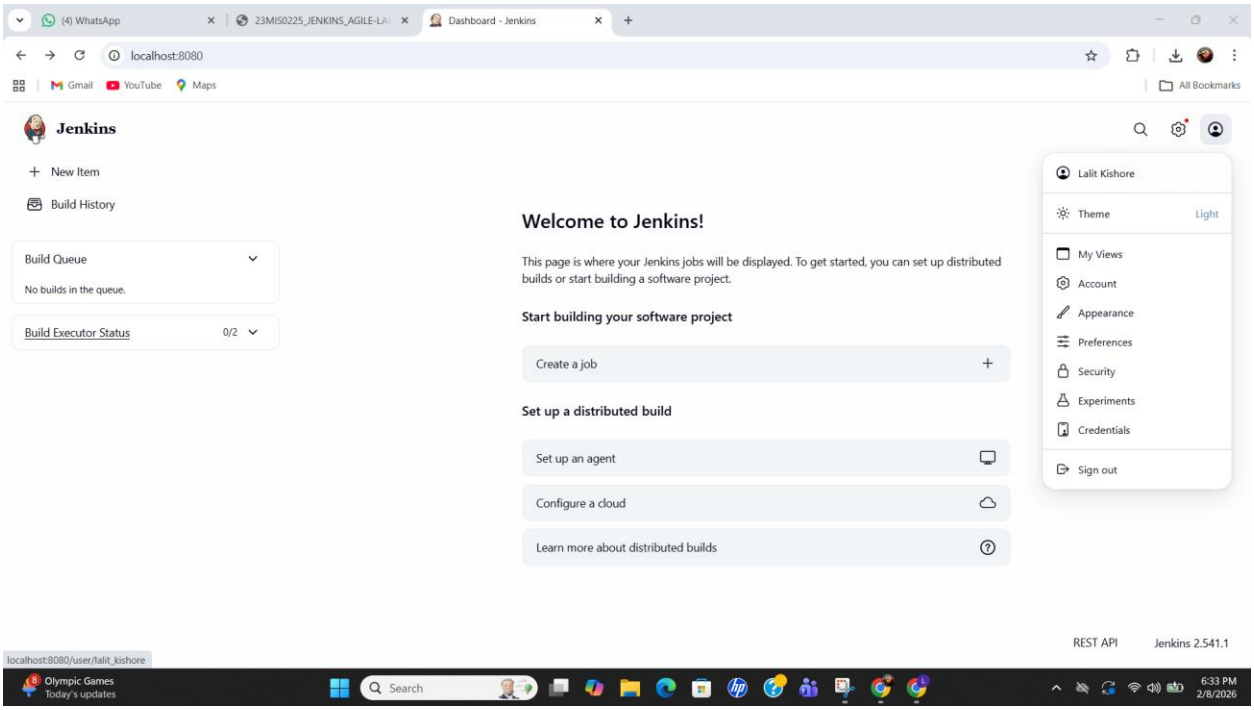
automatically delivered or deployed.

Real-world example:

When a developer pushes code to GitHub, Jenkins automatically builds and tests it instead of

doing it manually.

2. Install Jenkins (Windows / Linux)



3. Jenkins UI overview

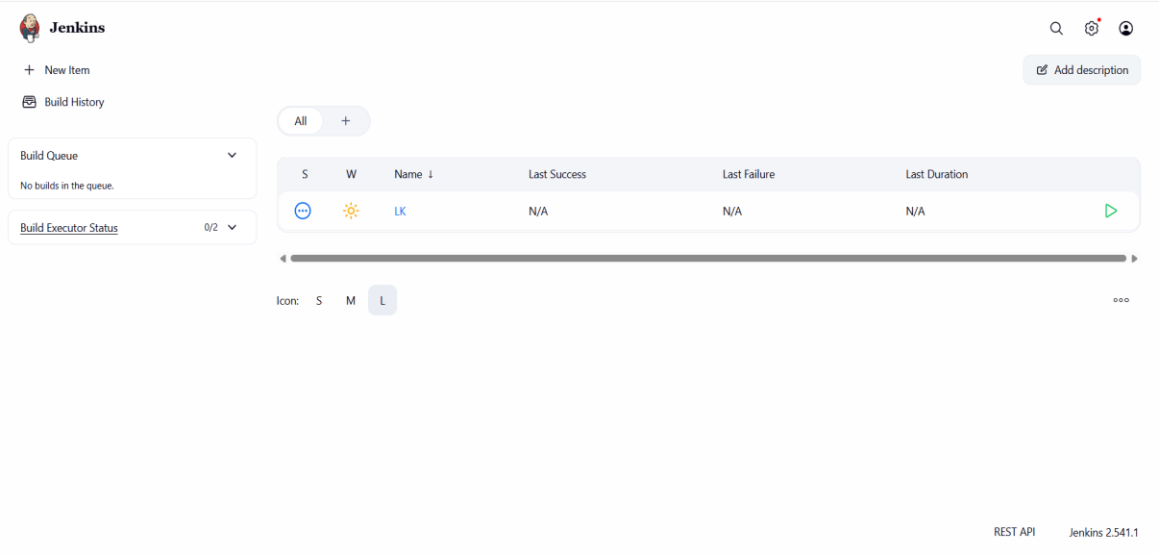
Dashboard: Shows all jobs

New Item: Create jobs

Build History: Shows past builds

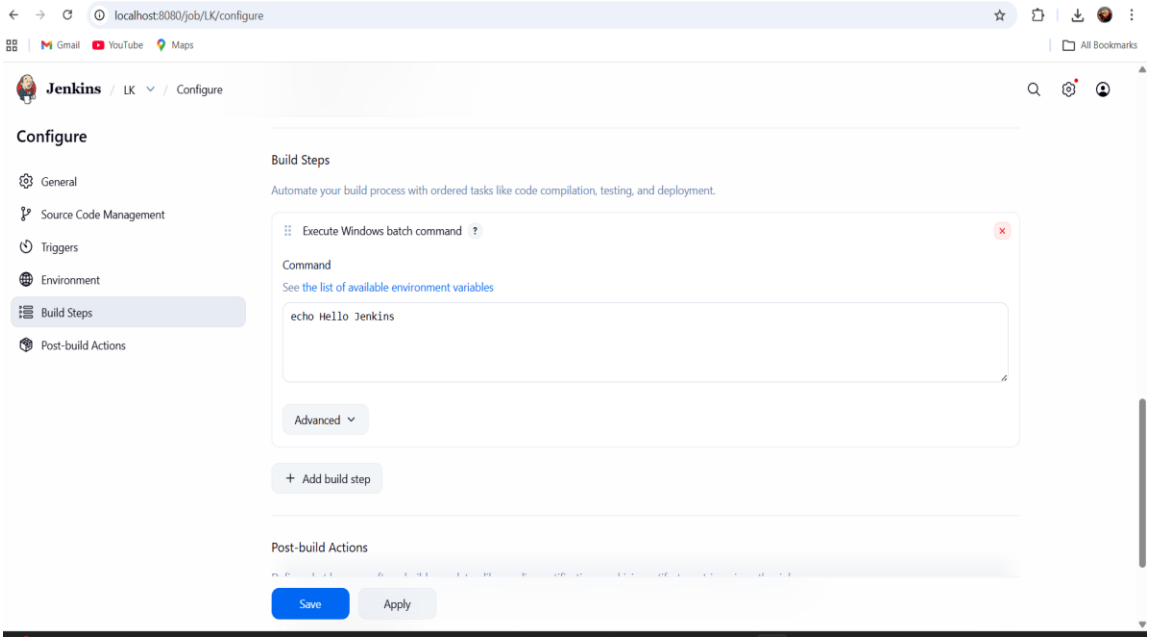
Manage Jenkins: Configure Jenkins settings

4. Create Freestyle Job

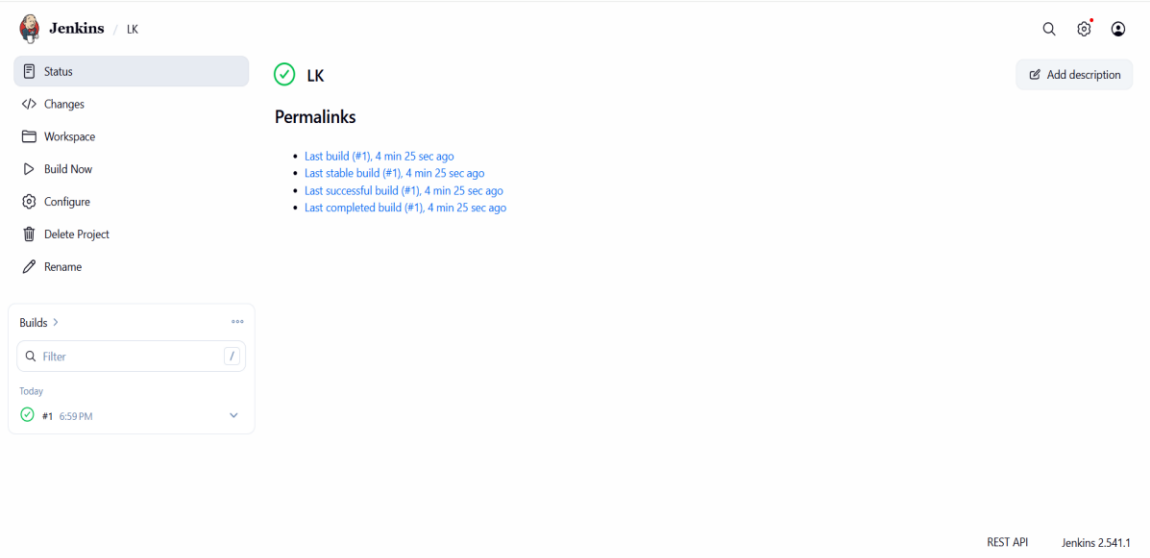


5. Add simple build step:

Windows: echo Hello Jenkins

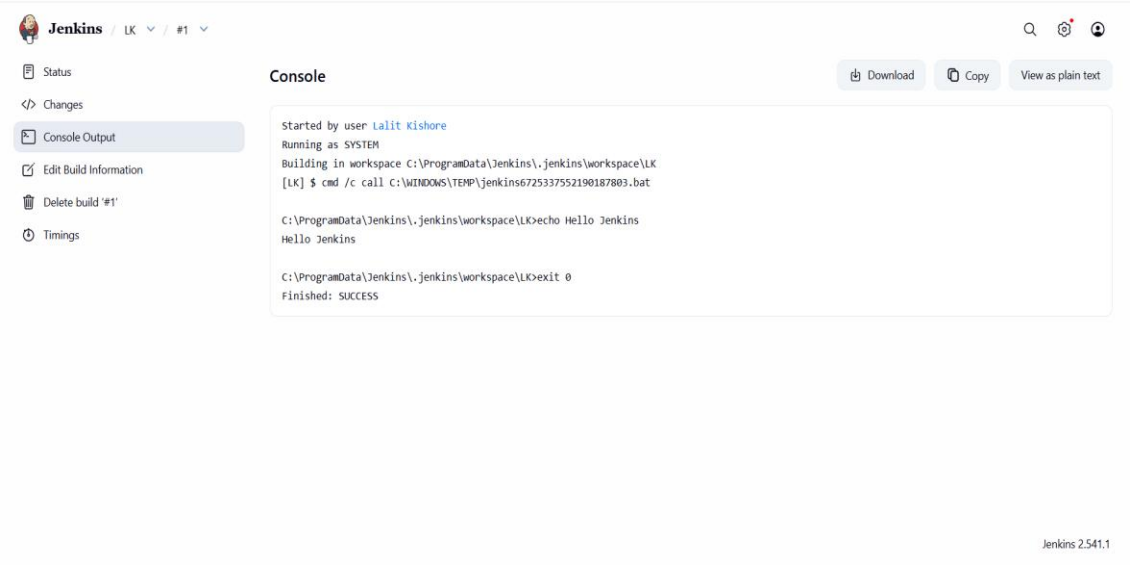


6. Run job manually

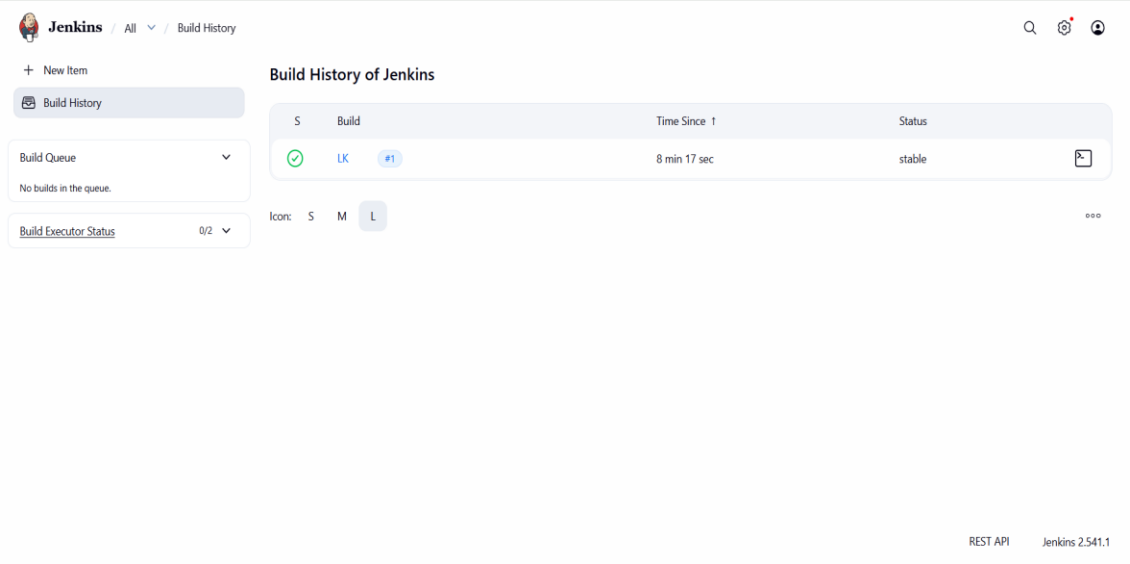


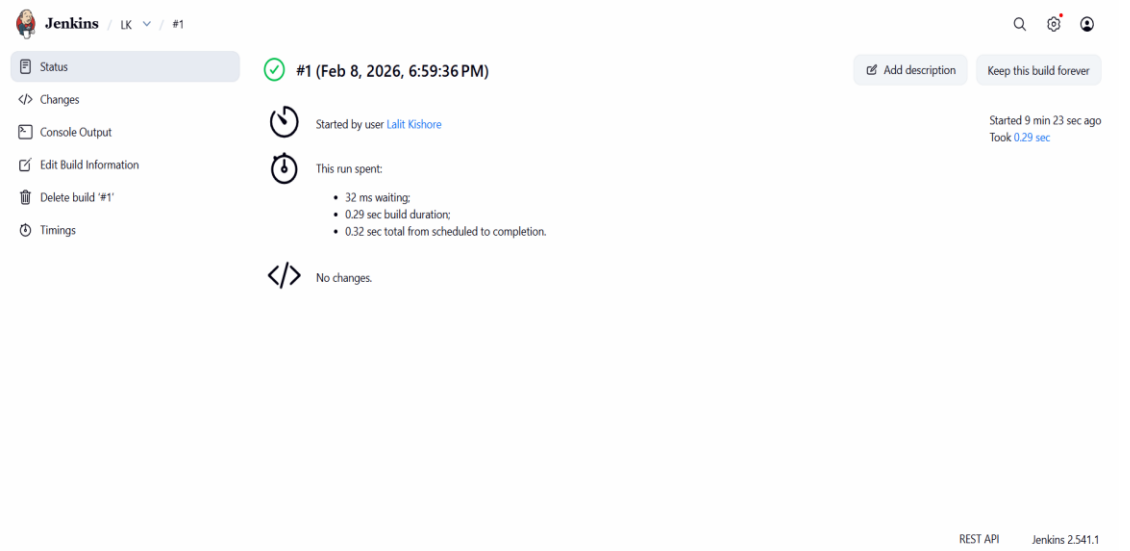
7. View:

Console Output



Build History





Outcome

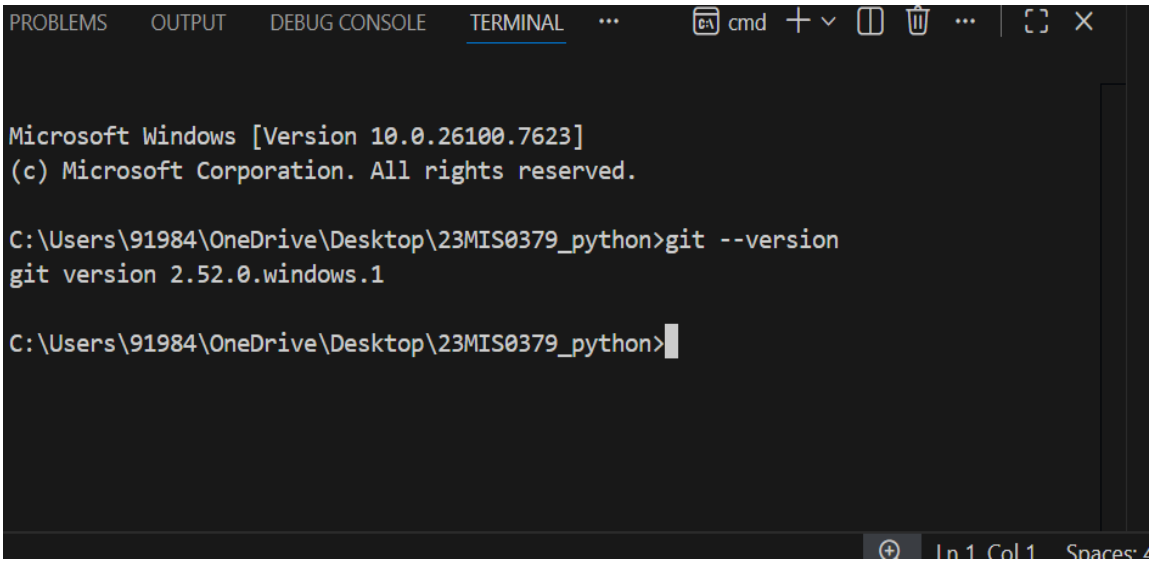
A learner can install Jenkins and run a basic job.

2. Learning Goal

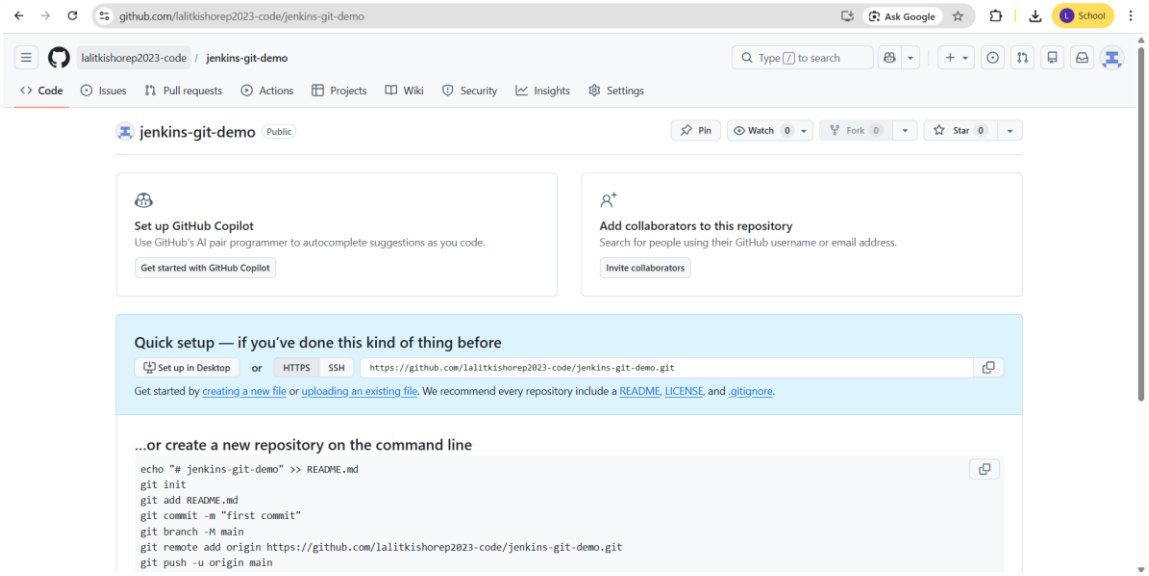
Work with source code and build automation

Stepwise Tasks

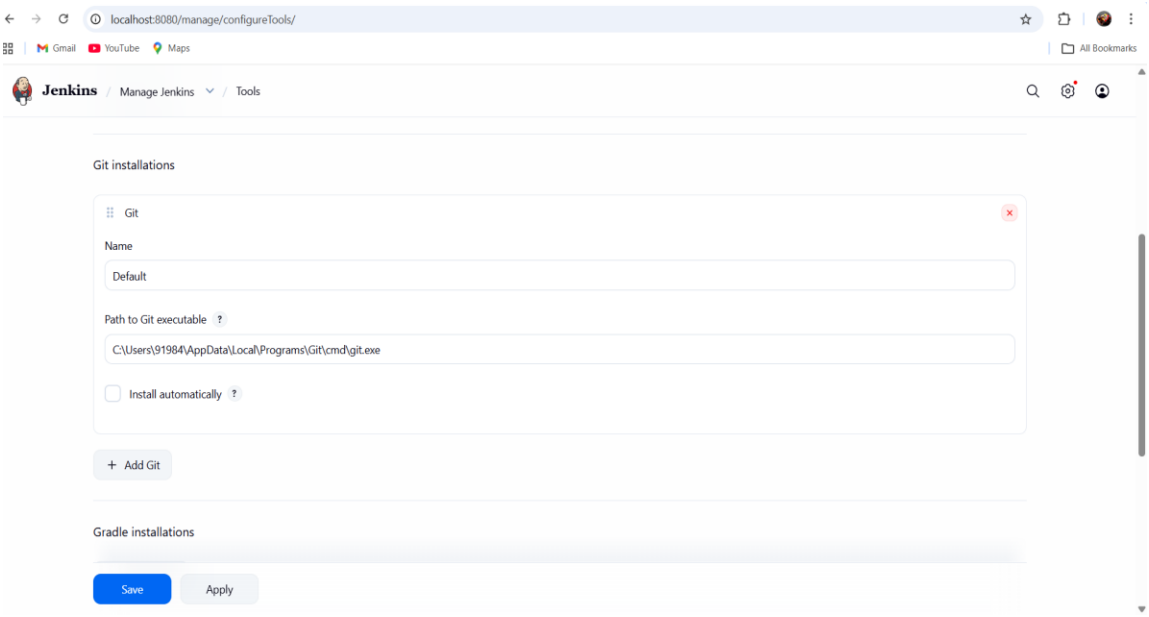
1. Install Git



2. Create GitHub repository

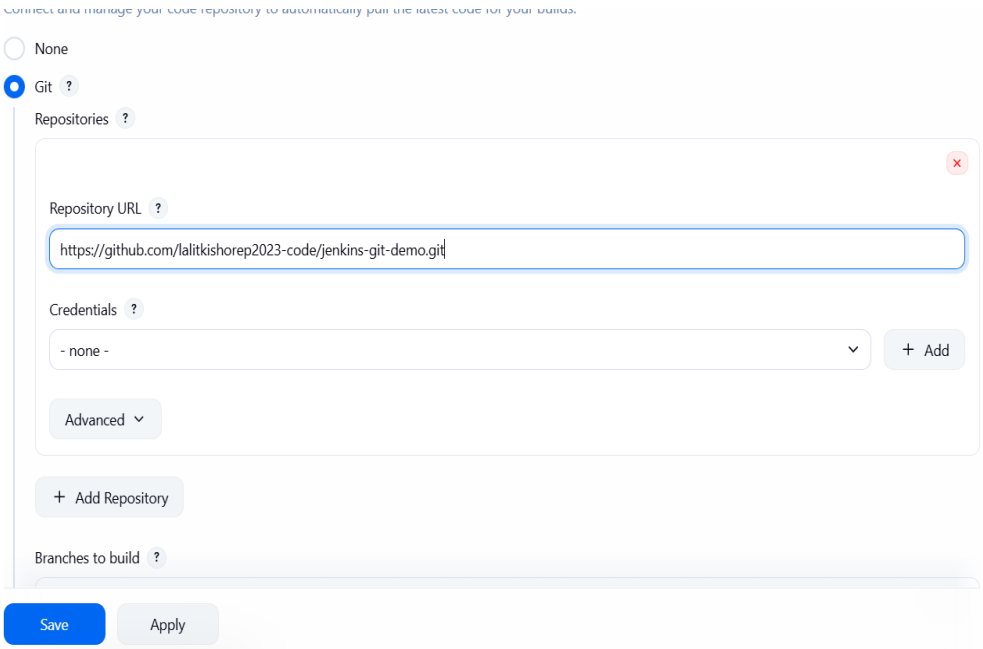


3. Configure Git in Jenkins



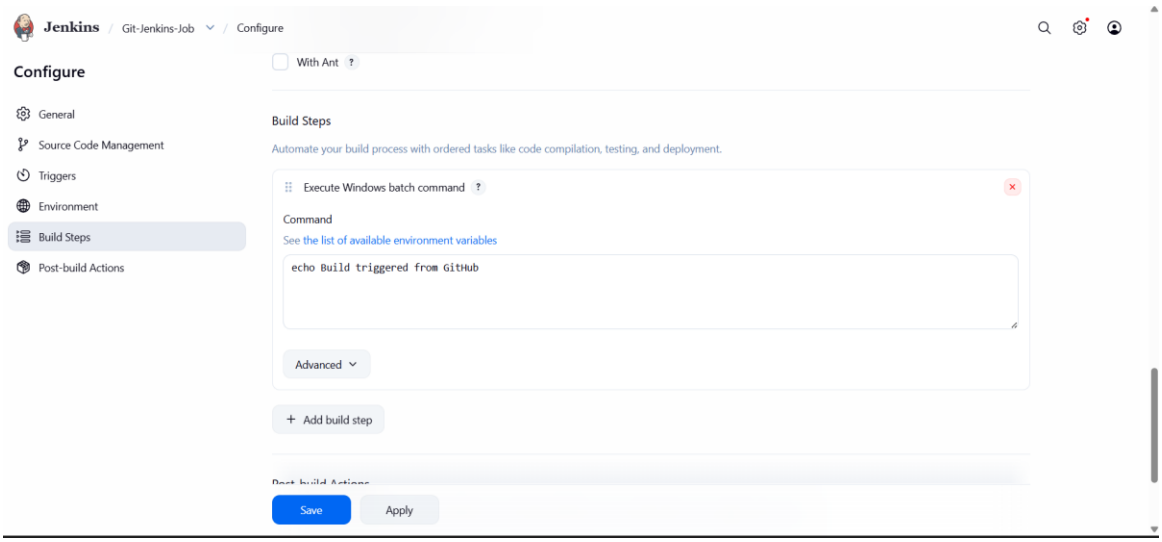
4. Create Freestyle Job:

Source Code Management → Git

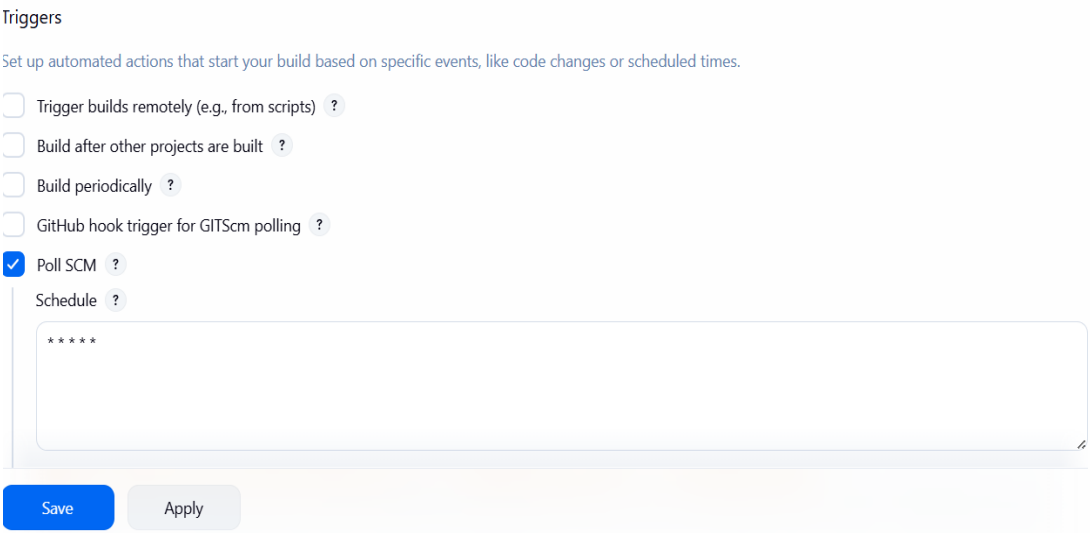


5. Add build step:

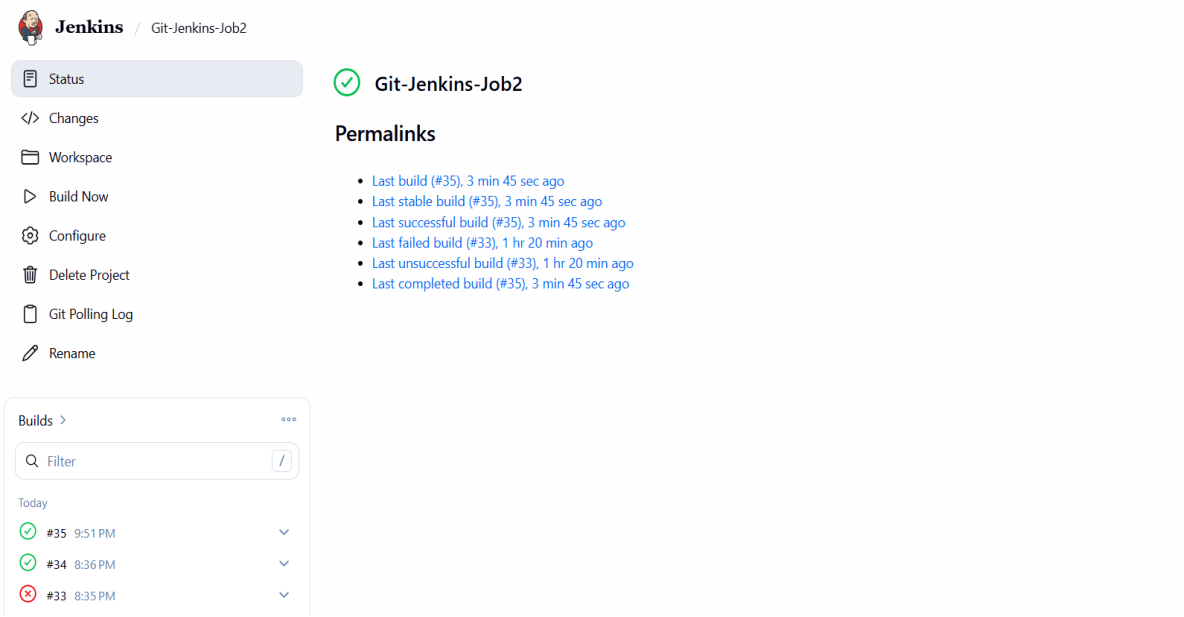
Windows batch / Shell

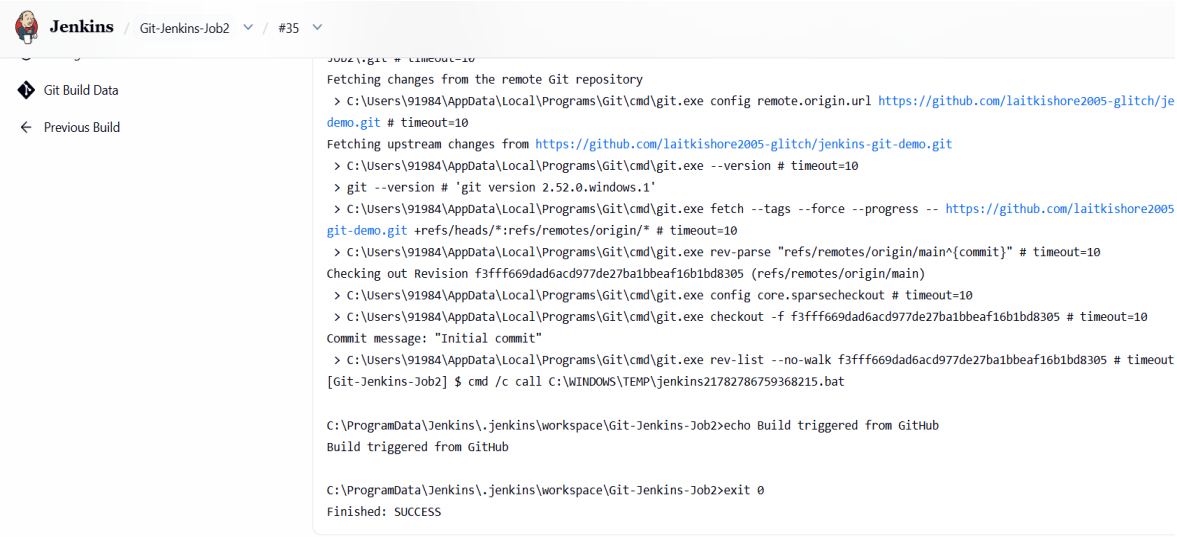


6. Poll SCM / Webhook trigger



7. Build automatically on code change





Outcome

Learner connects Jenkins with GitHub.

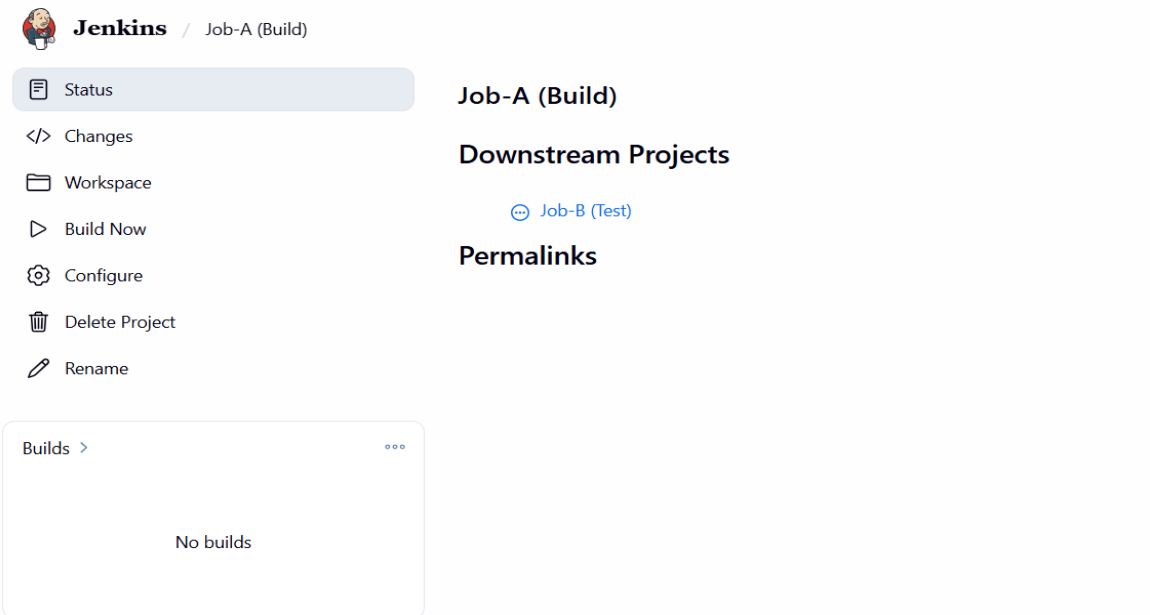
3. Learning Goal

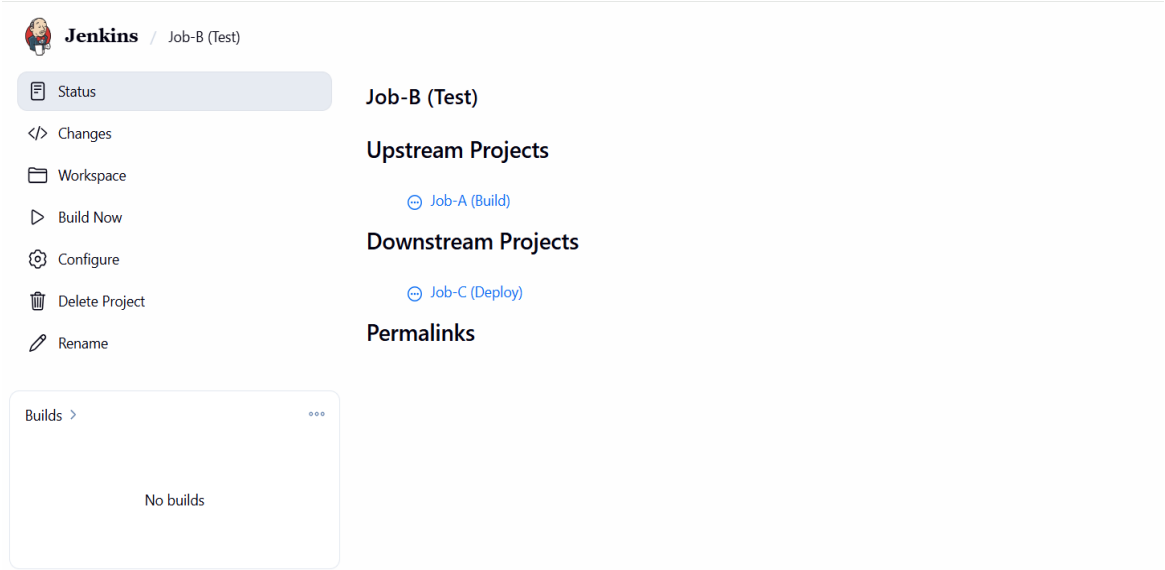
Understand job flow visually

Stepwise Tasks

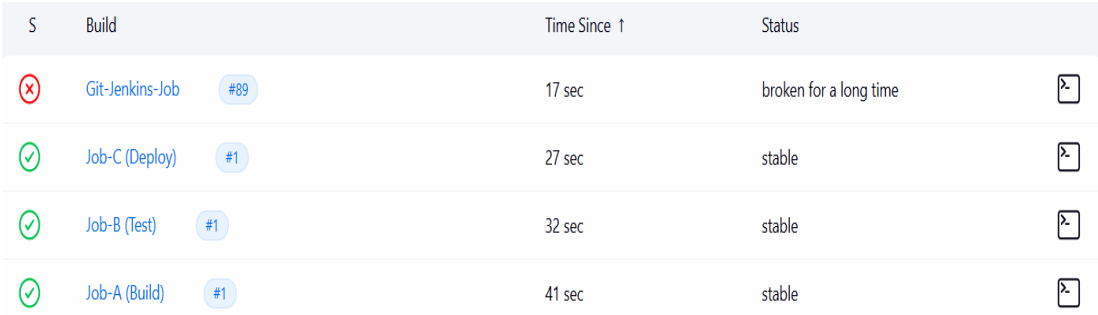
1. Create multiple Freestyle jobs:

2. Use: Post-build action → Trigger other projects

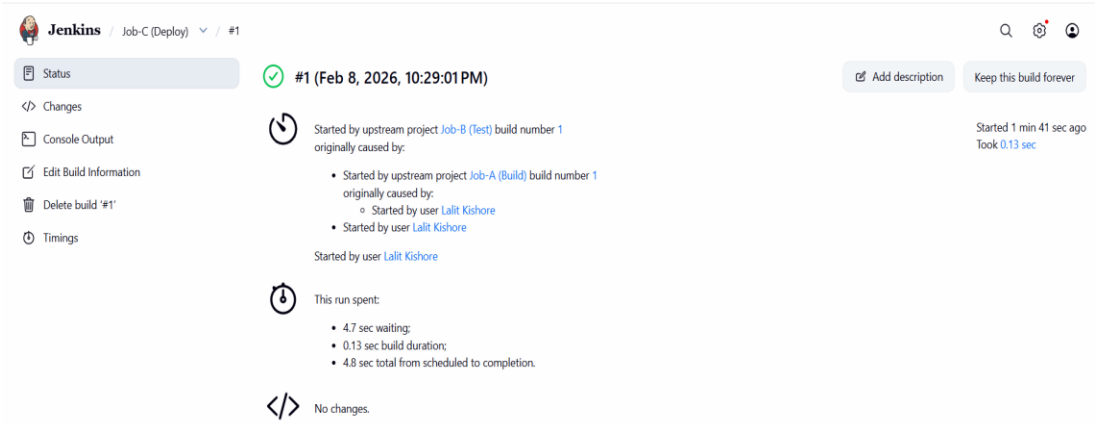




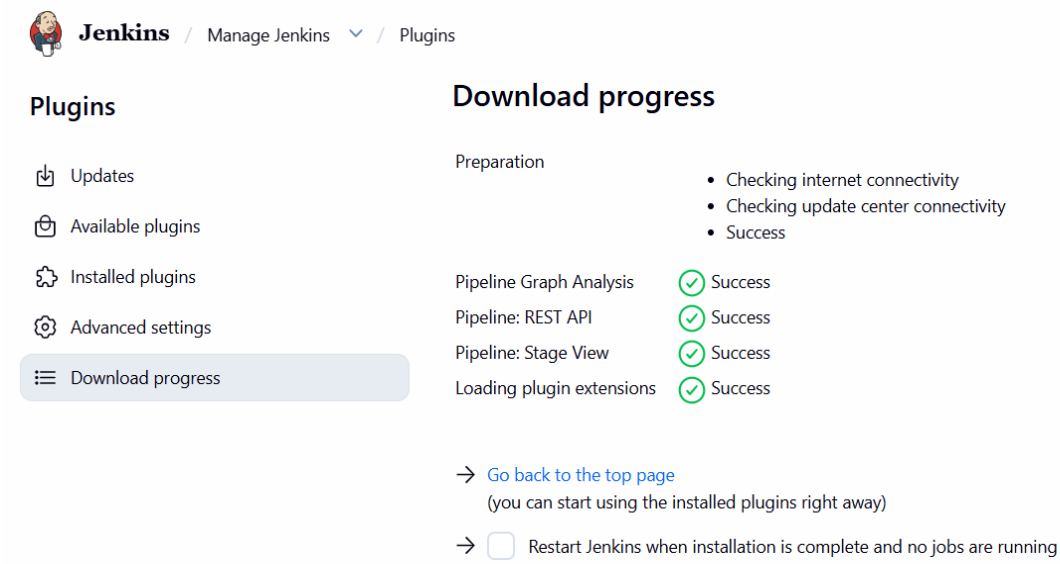
3. View job chaining visually



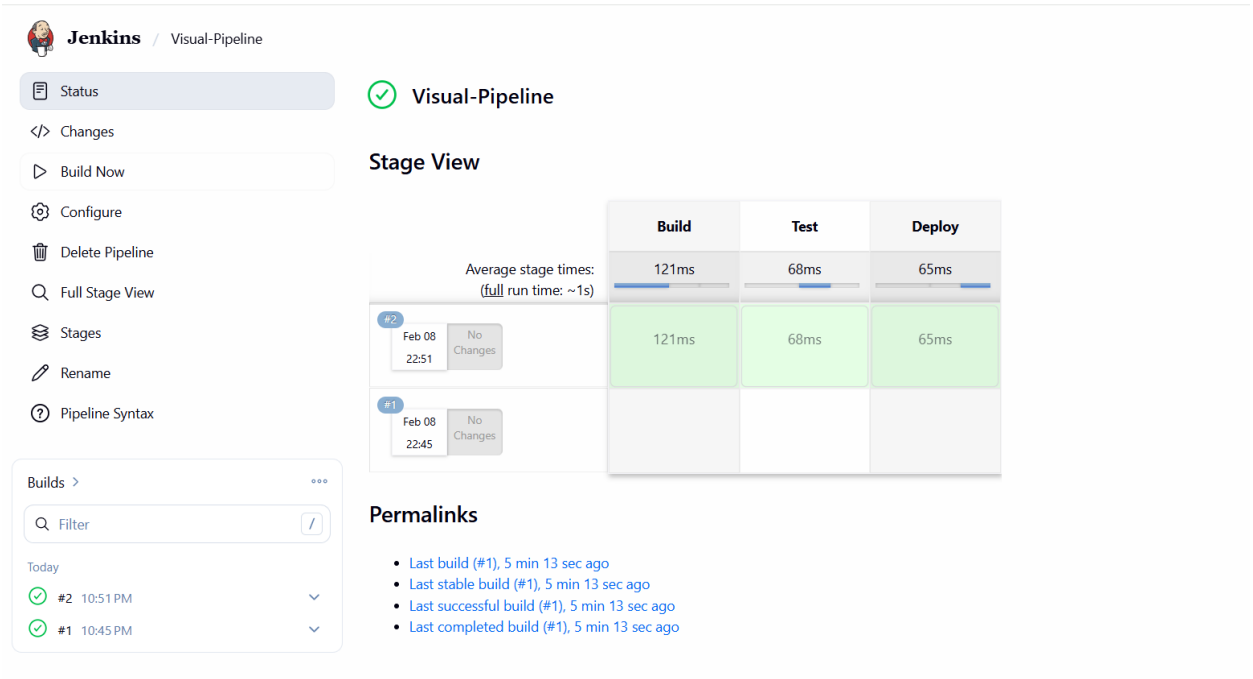
S	Build		Time Since ↑	Status
✗	Git-Jenkins-Job	#89	17 sec	broken for a long time
✓	Job-C (Deploy)	#1	27 sec	stable
✓	Job-B (Test)	#1	32 sec	stable
✓	Job-A (Build)	#1	41 sec	stable



4. Install Pipeline Stage View



5. Observe stage-wise execution



4. Learning Goal

Use Jenkins Pipeline & Jenkinsfile

Stepwise Tasks

1. What is Jenkins Pipeline?

Jenkins Pipeline is a set of automated steps written as code to define the entire CI/CD

process.

Instead of clicking many options in UI, we write:

i) Build
ii) Test
iii) Deploy

as code

Pipelines are:

i) Version-controlled
ii) Reusable
iii) Easy to automate

2. Scripted vs Declarative Pipeline

Scripted Pipeline

Written using Groovy code

a) More flexible

b) Complex syntax

c) Uses node { }

Declarative Pipeline

a) Simple, structured syntax
b) Easy to understand
c) Uses pipeline { }

3. Create Pipeline Job

LK	3 hr 59 min	#1	N/A	0.29 sec	
Visual-Pipeline	7 min 50 sec	#2	N/A	1.8 sec	

4. Write pipeline in Jenkins UI

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

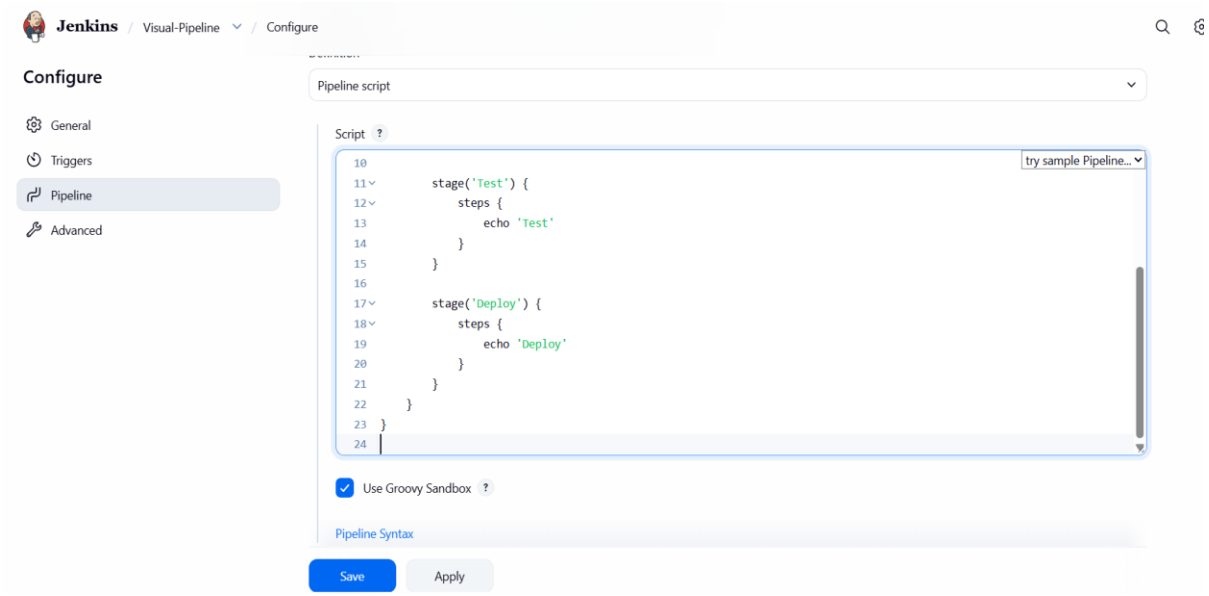
Pipeline script

5. Add stages:

i) Build

ii) Test

iii) Deploy



6. Introduce:

i) agent

ii) stages

iii) steps

agent

Defines where the pipeline runs

any → runs on any available node

stages

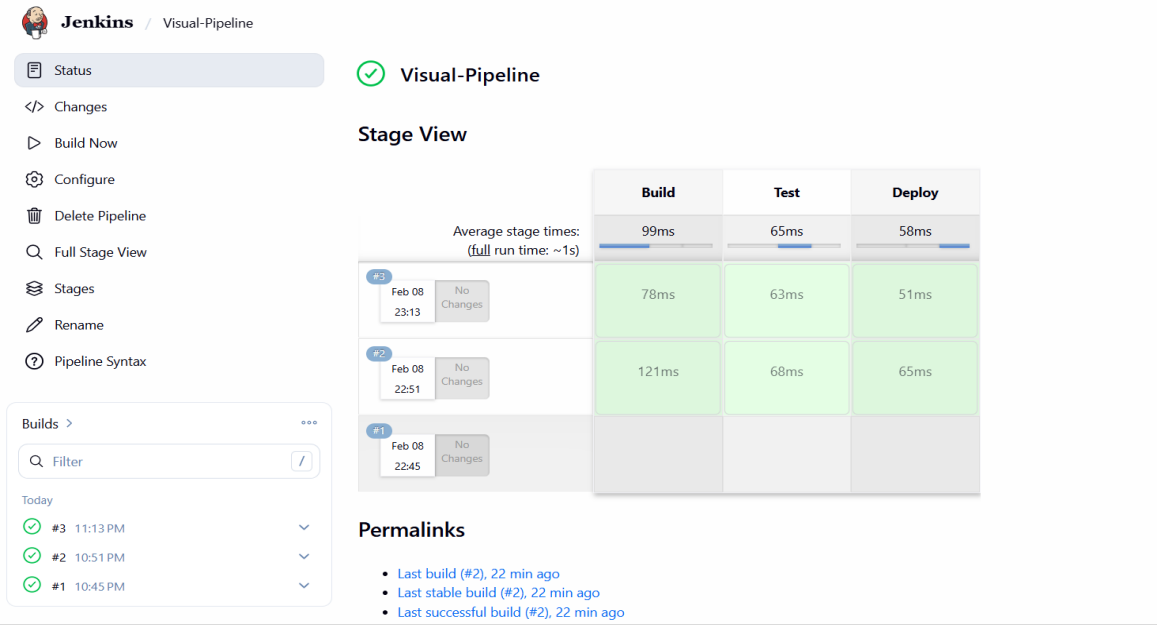
Group of all stages in pipeline

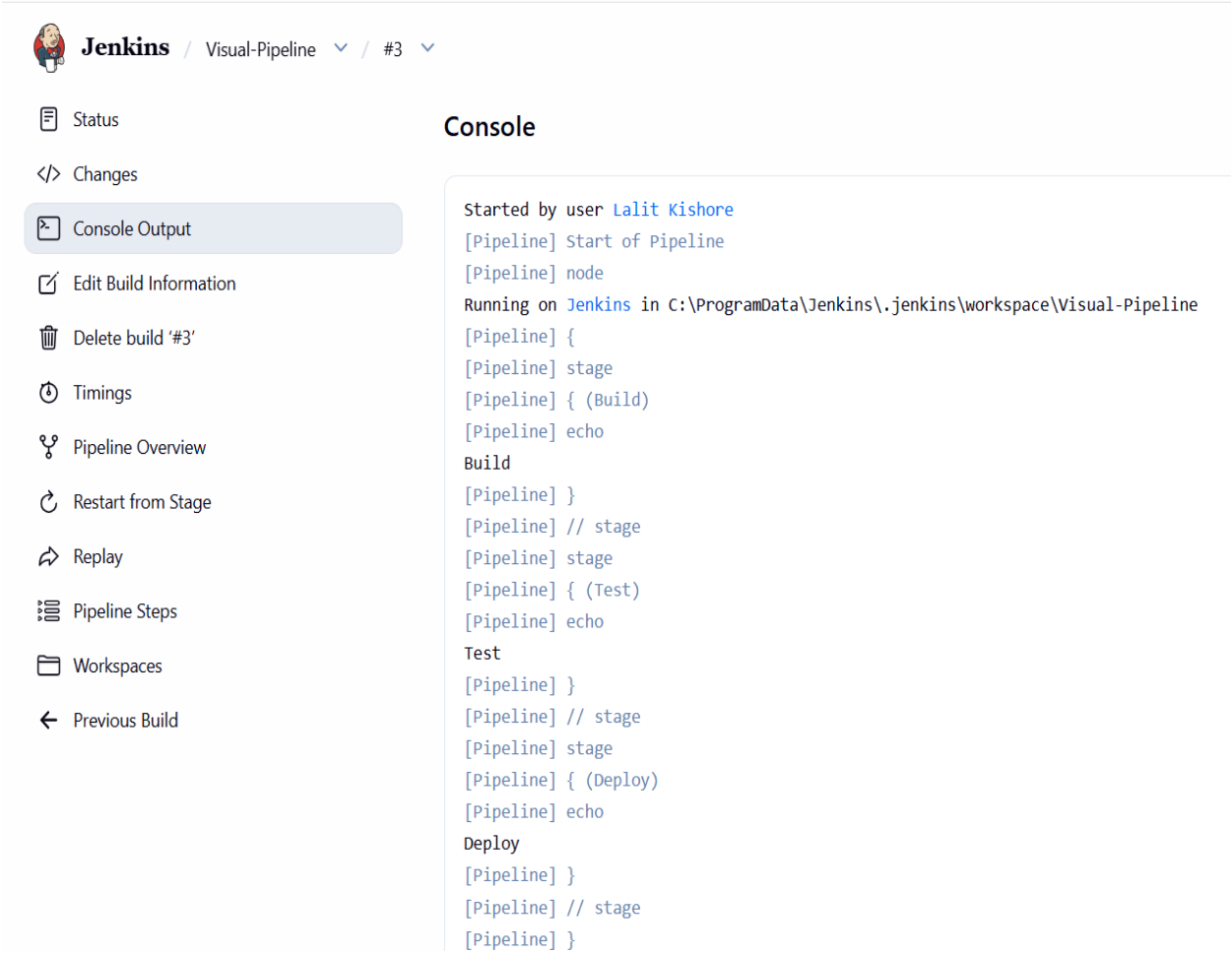
steps

Commands executed inside a stage

Sample Task

```
pipeline {
agent any
stages {
stage('Build') { steps { echo 'Build' } }
stage('Test')  { steps { echo 'Test' } }
stage('Deploy'){ steps { echo 'Deploy' } }
}
}
```





Outcome

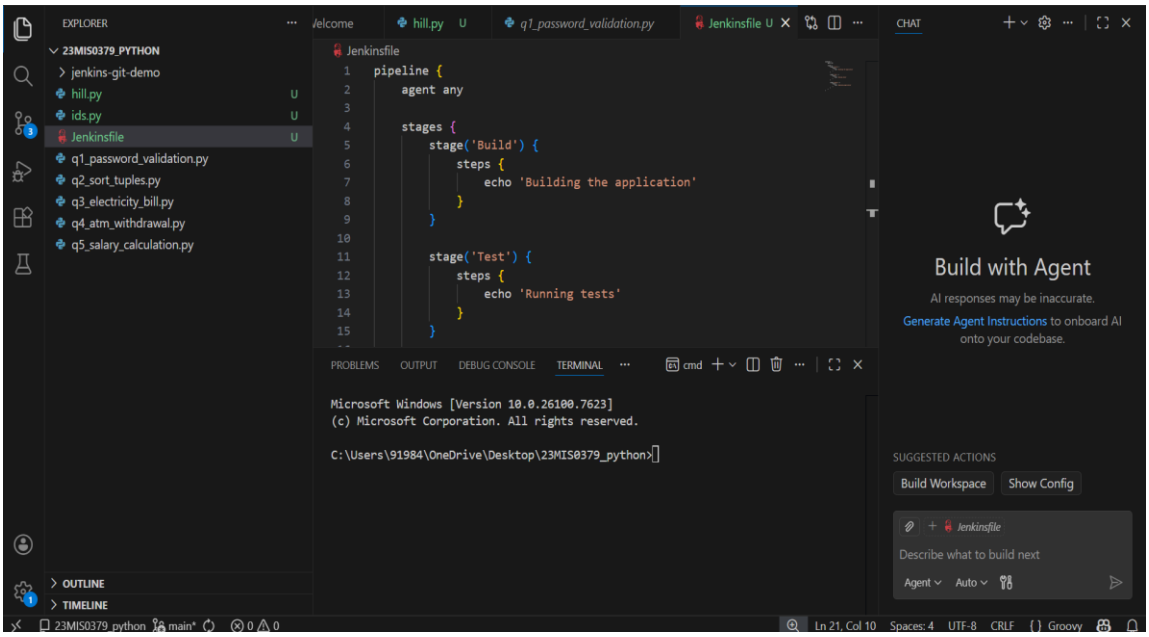
Learner writes basic pipelines confidently.

5. Learning Goal

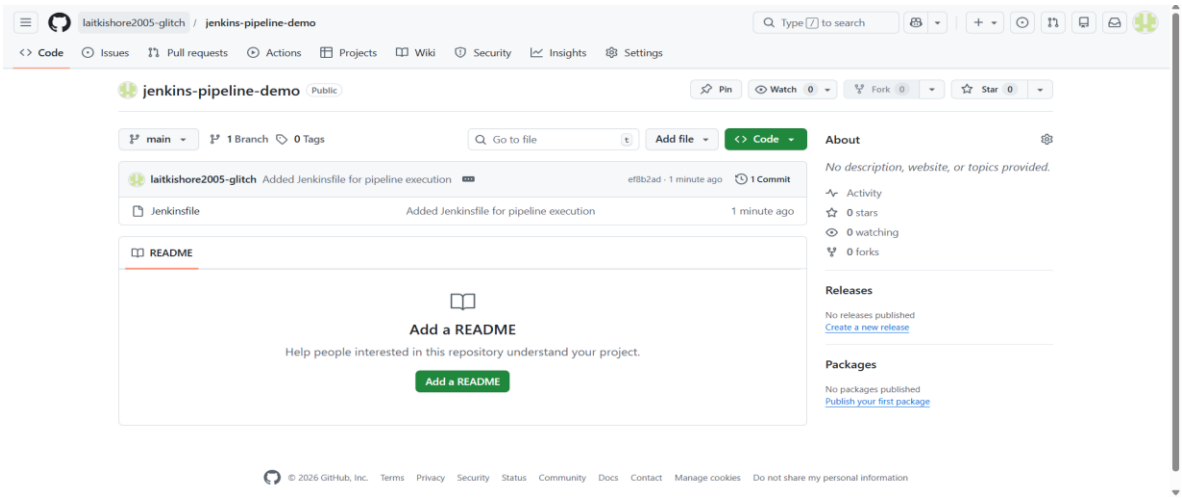
Pipeline as code from repository

Stepwise Tasks

1. Create Jenkinsfile

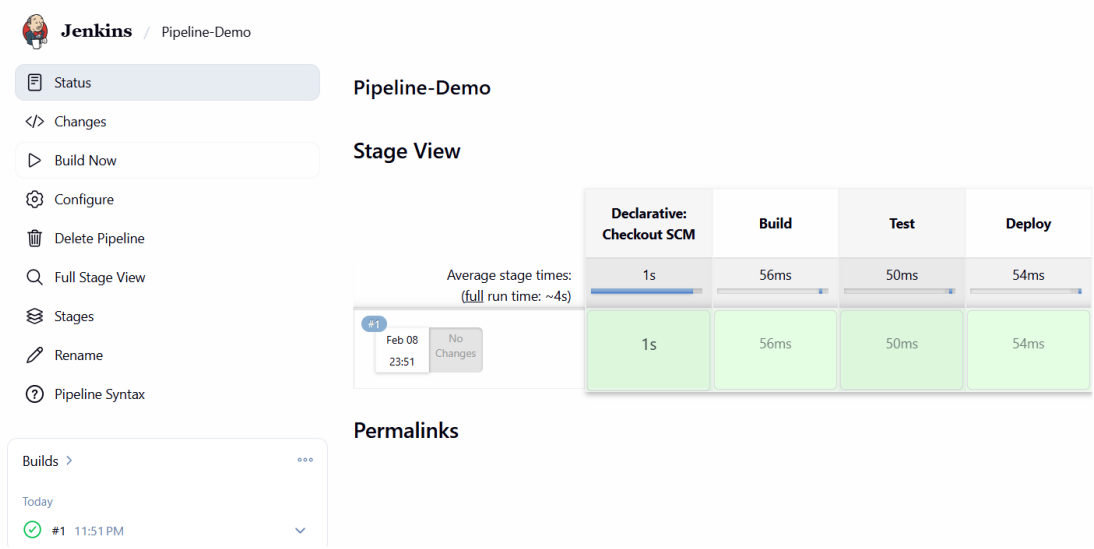


2. Commit Jenkinsfile to GitHub

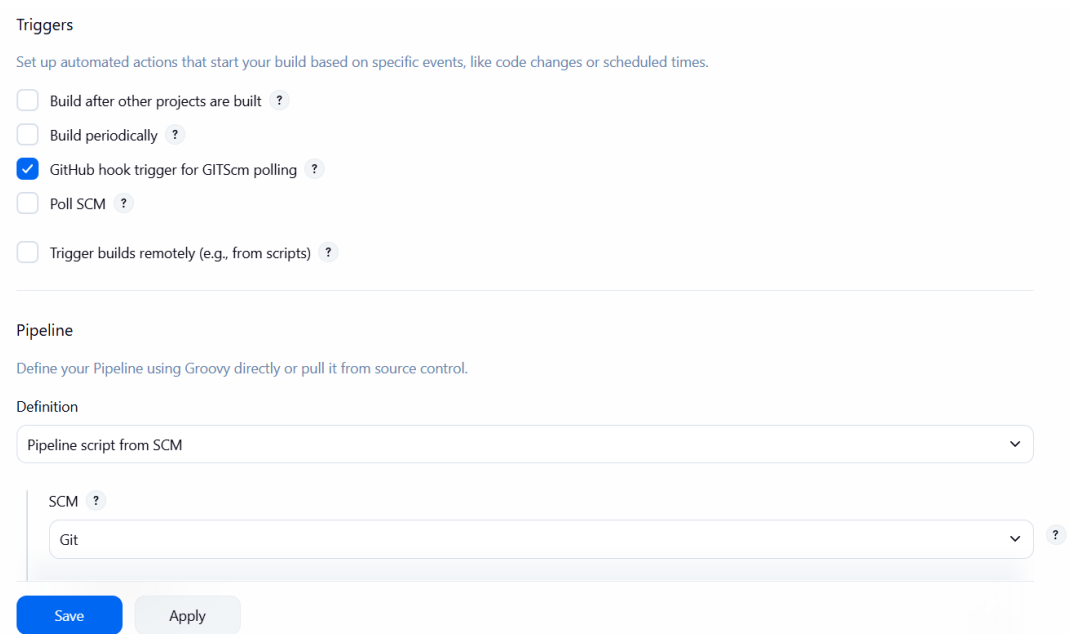


3. Configure Pipeline job:

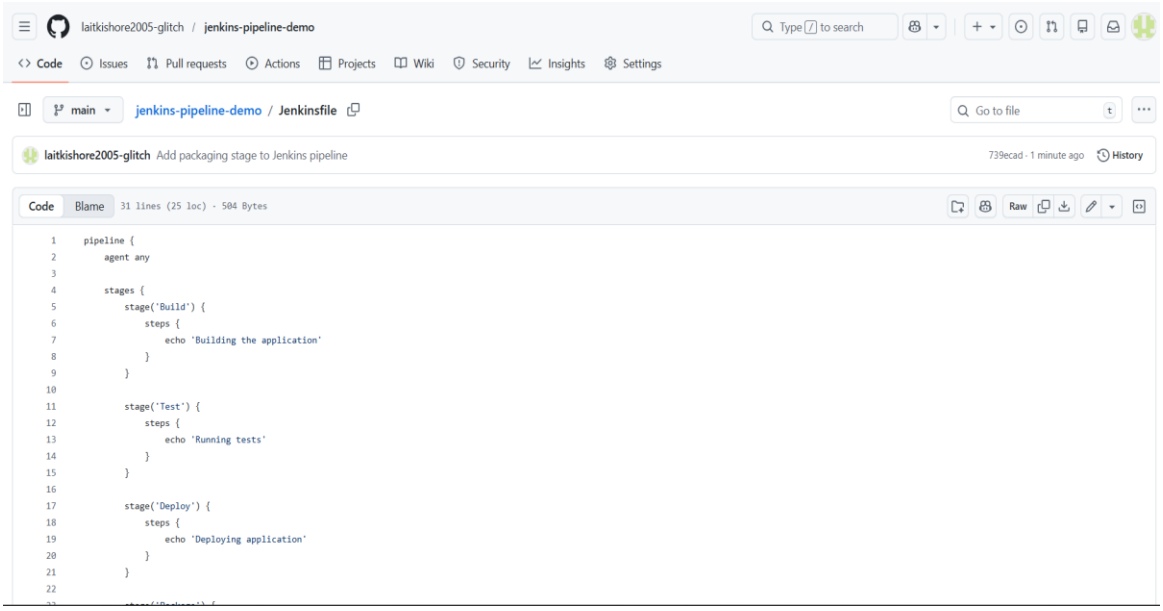
SCM → Git



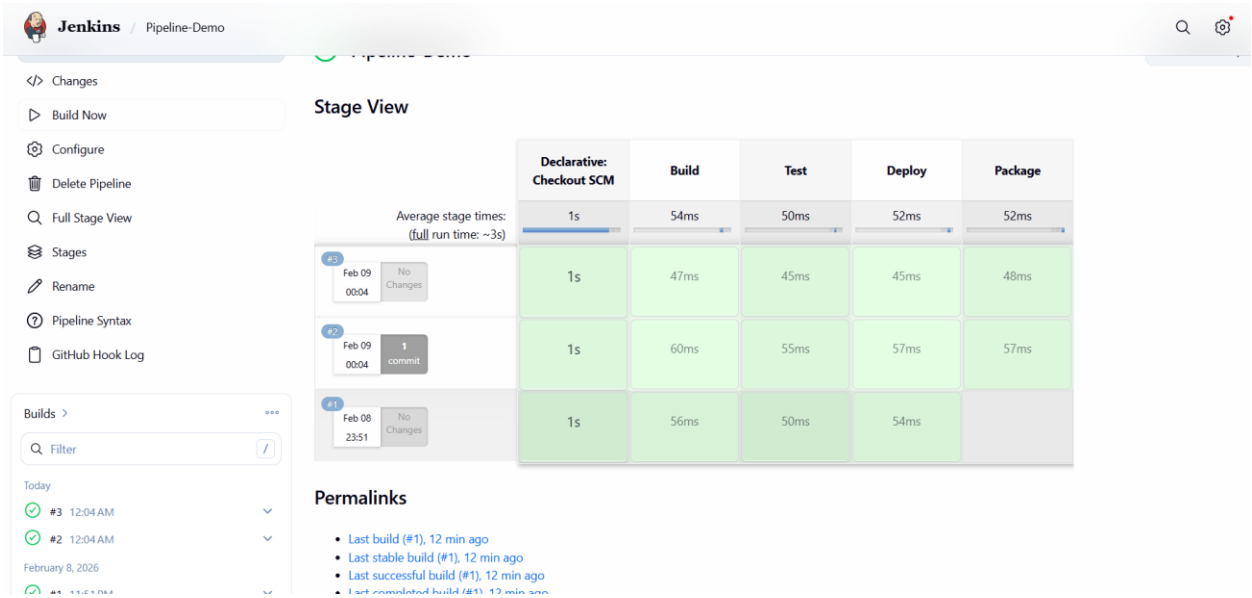
4. Trigger build automatically



5. Modify Jenkinsfile and rebuild



6. View Stage View



Outcome

Learner practices Infrastructure as Code concept.